

Learning Analytics in Action: A Mixed-Methods Study of Generative AI – ChatGPT’s Impact on L2 Academic Writing Development in English for Academic Purposes (EAP) Classrooms

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Abstract—In the field of English for Academic Purposes (EAP), the role of AI-assisted language learning in developing writing skills has been extensively researched and has yielded positive outcomes. Also, Learning Analytics (LA) can significantly contribute to comprehending and enhancing the responsible and reflective utilisation of AI tools in the educational context. However, its impact on language classroom education, particularly in EAP writing instruction, has not received widespread attention. This study addresses this gap by investigating the descriptions and explanations of preferred EAP writing learning strategies among a group of international students (N=42) engaged in EAP studies at a university, with a particular focus on their utilisation of generative AI-powered chatbot technology- ChatGPT. Results showed that EAP students believed generative AI technology to be user-friendly, simple to grasp, creative, and enjoyable for L2 writing learning. They may ask specific questions and receive immediate answers without waiting for responses or feedback. The quality of the written content they produce has improved significantly, and their opinion output is more novel and relevant, with varying degrees of improvement in grammar and syntax. The study's findings suggest that generative AI, such as ChatGPT, can facilitate the development of EAP writing proficiency and provide instant and constructive feedback within an informal digital language learning context. This research yields pedagogical implications for the development of L2 writing strategies for L2 students in EAP classes, utilising generative AI chatbots like ChatGPT. The findings could be a valuable reference for language researchers or teachers who intend to use generative AI-supported chatbots in EAP writing.

Keywords—*Learning Analytics, ChatGPT, Language assessment, English for Academic Purposes, EAP writing.*

I. INTRODUCTION

Generative artificial intelligence, including ChatGPT and numerous other text-generation tools, has profoundly transformed the modes and habits of learning in recent months. Due to the innovation in this technology, many students have increasingly relied on ChatGPT for studying and information retrieval [1]. Moreover, the integration of generative artificial intelligence and natural language processing (NLP) can produce exceptionally high-quality text and images, providing students with guidance and even assisting them in completing assignments [2][3]. The availability and accessibility of this

technology, particularly since ChatGPT's release by OPEN-AI in November 2022, have sparked unprecedented discussions on how teaching and learning in higher education should adapt and evolve [4][5].

In the realm of opportunities and challenges posed by these large language models in education, the discussions have become even more intense and extensive. The opportunities primarily focus on the potential role of these models as "assistants" in the creative process undertaken for learning purposes [6]. The central question is whether, when used cautiously by humans and effectively integrated with specific course content, they can enhance the efficiency of learning tasks. For example, they might facilitate scaffolding and self-assessment (such as providing feedback on one's writing) [7] or engage in content-rich conversations (practising spoken English) [8]. All these considerations not only pave the way for incorporating generative artificial intelligence as an auxiliary tool in learning task-oriented learning designs but also open possibilities for timely assessment through assignments providing feedback and personalized learning.

II. RESEARCH QUESTIONS

This paper explores these impacts in a very preliminary stage. It does so by examining data collected from students in English for Academic Purposes (EAP) language classes at a public university as part of the "Generative AI for learning English" initiative. Through amplifying the voices of EAP learners in higher education, the study investigates whether generative artificial intelligence, particularly ChatGPT (compared to other traditional chatbots), can effectively enhance the performance of EAP students learning English.

RQ1: What types of writing learning strategies are used by international L2 EAP students with the assistance of ChatGPT?

RQ2: To what extent may the generative AI - ChatGPT improve the EAP student's learning performance compared to the conventional AI chatbot approach in EAP writing?

III. METHODOLOGY

To assess the impact of the proposed approach, we conducted two writing walkthrough experiments in an EAP English course at a university in China. This study will conduct

a quasi-experiment to investigate the impact of the AI chatbot ChatGPT on the EAP learning of two groups of language class students within higher education. The chosen course aims to help students develop the skills to organise information and improve their English writing in the EAP classroom through learning and guidance based on a generative AI. Afterwards, semi-structured interviews are used to get further insightful ideas regarding using GenAI- ChatGPT. Ten participants are selected for further interviews. Specifically, what types of writing learning strategies are used and how they use them during the EAP writing were underscored.

Participant Recruitment and Randomization Process

The study was conducted at a private language training institute in Eastern China, anonymized as "LTC Institute," which specializes in English for Academic Purposes (EAP) programs. This institution was selected due to its standardized EAP curriculum (CEFR B2-C1 level), stable class sizes (20–25 students per class), and openness to integrating experimental technologies. Recruitment began by distributing electronic invitations to 120 students enrolled in two parallel EAP classes during the Fall 2024 semester. Eligibility criteria required participants to be full-time undergraduates aged 18 or older, enrolled in a 16-week EAP writing course, and lacking formal training in AI writing tools. English majors and students with prior ChatGPT experience (over three months) were excluded to minimize bias. A bilingual questionnaire (Chinese/English) was administered via the institution's Learning Management System (LMS), covering informed consent, demographics, AI tool usage habits, and participation intent. Within 72 hours, 87 responses were received (72.5% response rate), with 52 students provisionally agreeing to join the study.

To ensure balanced group allocation, a stratified block randomization method was employed. Participants were stratified by CET-4 scores (≤ 550 vs. > 550), gender (male/female), and major (STEM vs. humanities). Using block randomization (four-person blocks: two experimental, two control), participants were randomly assigned to either the experimental group ($n=22$), which used ChatGPT for writing feedback, or the control group ($n=20$), which received traditional teacher-led instruction. A waitlist of 10 students was prepared to replace potential dropouts. Teachers were informed of group assignments to implement interventions, while students were only told they would use "different writing support tools." A phased incentive system (30% after pre-tests, 70% after post-tests) and weekly SMS reminders ensured full retention. Baseline equivalence between groups was confirmed statistically, with no significant differences in age, gender distribution, CET-4 scores, or major categories (all $p > 0.05$).

After the intervention, 10 participants from the experimental group were selected for qualitative interviews. The qualitative interviews will be analysed using thematic analysis [9]. Five were chosen based on significant writing improvement (top 30%), two each from high, mid, and low CET-4 score tiers, and one volunteer. Ethical measures included post-study ChatGPT workshops for the control group, AI-redacted audio storage on encrypted servers, and a waitlist replacement protocol for three students who withdrew post-pre-test. This process adhered to methodological rigour, balancing

randomization transparency with contextual adaptability for educational research in China's private language training sector.

IV. RESULTS AND DISCUSSION

Regarding research question 1, To generate ideas from the data, I used thematic analysis, a qualitative approach for identifying and interpreting patterns [9], which involves six steps:

1. FAMILIARIZATION WITH THE DATA: The first step involves reading and re-reading the data to become deeply familiar with its content. This helps the researcher immerse themselves in the data, noticing key patterns, ideas, and initial thoughts. It includes noting initial ideas and making sense of the content to prepare for the analysis process.
2. GENERATING CODES: Interesting features were highlighted
3. FIND THEMES: Codes were grouped and organised into broader themes.
4. REVIEW THEMES: I reviewed the themes for accuracy.
5. DEFINE THEMES: I named themes and created tables to represent them with selected examples from the data as evidence (**Appendix A**).
6. REPORT: I wrote the analysis using data evidence.

Therefore, three themes were found by analysing the qualitative interview data. Students can interact with ChatGPT to generate ideas for their writing assignments. By asking open-ended questions, students can receive prompts, suggestions, and directions for developing a coherent structure or finding relevant sources. Also, in terms of prompting for examples and arguments:

"ChatGPT can help students brainstorm arguments or examples related to their essay topics. This is particularly useful for students who struggle with organizing their thoughts or articulating their ideas clearly."

The results of this interview are almost identical to those of Baidoo-Anu et al. [2] and Bozkurt et al. [3], showing that GenAI ChatGPT can help students complete assignment writing tasks (content) and that brainstorming at the beginning of the process is also very useful. For outlining and structuring ChatGPT can assist students in creating outlines by suggesting logical structures for their writing. For example, students can ask ChatGPT to provide a general essay structure (introduction, body paragraphs, conclusion), along with tips on organizing ideas effectively. Also, in terms of Organizing paragraphs, Students can use ChatGPT to help them organize their paragraphs and develop cohesive connections between ideas [10] [11]. The tool can suggest how to structure paragraph openings and ensure that each section flows logically.

For Grammar and Syntax Correction, Gen AI ChatGPT can help students by pointing out grammatical mistakes (e.g., subject-verb agreement, tense usage) and suggesting corrections, allowing them to learn from their errors. These findings align well with Liang et al. [12], research on Gen AI

Chatbot. In his study, he suggests a synchronous conversational corrective feedback (CCF) approach that offers metalinguistic explanations (ME) and self-correction. Also, he concluded that the Gen AI chatbot feedback is more efficient than other tools, especially in L2 writing grammar error feedback. Regarding the improvement of sentence structure, students can ask ChatGPT to rephrase sentences for clarity or to make their writing more complex and varied. This is particularly beneficial for improving the fluency and sophistication of academic writing.

Regarding research question 2, Table 1 presents the mean scores and standard deviations of the two groups of students' writing performance before and after the test. The analysis reveals significant differences between the experimental and control groups in the final writing scores, content, outline writing ideas, grammar and syntax. We can see that the overall EAP writing score was improved (The post-test Mean is 25.86 score in EAP writing). However, no significant differences (Pre-test Mean is 22.58 VS post-test Mean is 23.08) were found in scores related to logic and outline structure. This indicates that the writing instruction method based on large language models substantially positively impacts students' writing performance in terms of overall scores, content, and language/grammar. The ANCOVA results focused on group-level comparisons rather than individual group p-values, as the analysis evaluates intergroup differences holistically. The experimental group exhibited statistically higher adjusted means than the control group ($p < 0.05$). Notably, post-hoc analyses of *content quality*, *outline/structure*, and *grammar/syntactic complexity* indicated that ChatGPT integration led to both improved writing outcomes and time efficiency gains for learners.

TABLE I. ANCOVA RESULTS OF WRITING PERFORMANCE

Variance	Group	N	Pre-test		Post-test		Adjusted Means	SE
			Mean	SD	Mean	SD		
Overall Writing scores	EG	22	23.04	0.71	25.86	0.43	25.86	0.07
	CG	20	23.04	0.59	24.08	0.44	24.50	0.07
Content	EG	22	24.58	0.81	25.58	0.56	25.68	0.90
	CG	20	23.08	0.71	25.85	0.47	24.45	0.90
Outline/Structure	EG	22	22.58	0.76	23.08	0.58	23.18	0.09
	CG	20	22.25	0.79	23.15	0.48	23.36	0.09
Grammar & Syntax	EG	22	22.05	0.82	23.98	0.56	24.08	0.09
	CG	20	22.98	0.78	24.78	0.52	24.98	0.09

The findings from the qualitative data further support the quantitative results. In the semi-structured interview questions, students primarily noted improvements in their ability to argue points, such as providing examples and using language, highlighting the significant positive impact of ChatGPT. In terms of content, students deepened their understanding of EAP writing topics through repeated interaction with the tool. After generating relevant ideas based on the rich information and materials provided by ChatGPT, their learning efficiency and output were greatly accelerated.

Some participants also shared their perceptions of utilising the ChatGPT, most participants are happy about the affordances of this GenAI tool. For example,

“ In conversations with the generative AI chatbot-ChatGPT, I became more focused and became more interested in EAP writing, finding it easier to immerse myself in the learning tasks. With its help, I gradually began to enjoy EAP writing. Essentially, this GenAI is user-friendly, simple to grasp, and creative. I even realized for the first time that this kind of companionship made EAP writing feel less difficult—at least, I no longer resist it.” (P3)

Also, as P7 mentioned:

“I had heard about generative AI from my classmates but had never experienced it myself. However, one day after class, everyone was discussing how helpful GenAI was for learning EAP writing, which sparked my interest. After trying out the large language model to assist with all my writing tasks in this course, I found it extremely convenient. I look forward to the emergence of even more advanced AI tools that can further improve writing in the future.” (P7)

Our research study provides insights into the development and future of AI in the field of EAP writing. However, it is important to emphasize the impact of prior knowledge on these findings. Participants in this study were provided with basic knowledge of ChatGPT and the EAP writing topics before the experiment began, which gave them more opportunities to identify incorrect prompts or make more effective use of ChatGPT for independent learning, thereby building confidence. For research and applications involving the basic use of ChatGPT, it may be beneficial to offer additional training or reminders before usage, particularly for groups with limited prior knowledge, such as lower-year students or beginners with poor English proficiency.

V. CONCLUSION AND FUTURE RESEARCH

This study, as an initial exploratory research endeavour, offers a reference for understanding the learning analytics of generative artificial intelligence (AI) in the context of academic writing for English for Academic Purposes (EAP) students in higher education. Based on the findings of this quasi-experimental study, we establish a foundational platform for further investigation and engagement with EAP educators, providing directional insights for pedagogical reform. The research underscores the imperative for the evolution and

progression of learning design and instructional approaches, subsequently guiding learning analytics to adapt correspondingly to these generative AI tools. It emphasizes the necessity for analytics to discern between AI-assisted actions and human intelligence, aligning with pedagogical intentions. In addition to offering developers, educational policymakers, educators, and researchers useful advice on how to successfully use ChatGPT in EAP writing learning while preserving teaching quality, these findings expand our knowledge of EAP students' adoption of GenAI technology. In the future, more research is needed to provide teachers with instructional feedback by studying the impact of generative AI on learners. This can enhance the role of teachers and facilitate their efforts in implementing pedagogical reforms. Additionally, EAP in higher education requires a deeper understanding of analytical techniques that are more suited to analyzing student interactions with generative artificial intelligence. Ethical tensions should also be addressed in the future.

Appendix A: Coding example table

Coding Scheme	Interview transcript example
Enhance the autonomy	finding it easier to immerse myself in the learning tasks. With its help, I gradually began to enjoy EAP writing
User- friendly	The GenAI chatgpt is just very easy to operate, I can use this very easily.
Creative	I can use ChatGPT to generate the example, and to generate the brainstorm for my EAP writing as well, whenever I have no example to write in my assignment, I will prompt ChatGPT and ask it to give me 2 -3 examples in my assignment.
Enjoyable in L2 writing	I feel no longer lonely and depressed. I start to fall in love in writing my L2 writing.
Use for inquiry	I can use ChatGPT to ask some basic examples in my writing. Sometimes I know that my questions are very stupid. So I feel ashamed that I may not ask my teachers or students.
Relatedness/peers	“I had heard about generative AI from my classmates…….”

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